

Degree Project in Materials and Process Design

First cycle, 15 credits

Ab initio modelling of magnetic properties of Fe-Cr-Ni duplex stainless steels

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Abstract

As duplex steels have become increasingly widespread in recent years, it has become essential to be able to characterize these materials easily and accurately. This study aims to simplify and diversify the characterization process by establishing relationships between the magnetic properties of the Fe-Cr-Ni ternary alloy and its composition. To this end, the magnetization and Curie temperature of the ferritic phase are calculated numerically using ab initio methods. Firstly, the ground-state energy and magnetization are computed using the EMTO (exact muffin-tin orbitals) method. This provides an initial estimate of the Curie temperature using mean-field theory. Next, the exchange interactions between atoms are computed and used to construct the Heisenberg Hamiltonian required for the Monte Carlo method. These simulations make it possible to calculate the magnetization at finite temperatures and thus to determine the Curie temperature in a second way. The results show that magnetization decreases with the addition of chromium and nickel. The trend is nearly linear and is therefore well approximated by a plane. The values obtained are consistent with those reported in the literature. The two different approaches to determining the Curie temperature yield fairly similar results, showing a significant decrease in the Curie temperature with the addition of alloying elements. Surprisingly, chromium and nickel have nearly identical effects. The results are quite far from the experimental values, highlighting the limitations of Monte Carlo simulations as performed in this context. However, these results demonstrate the promising potential of characterizing duplex steels based on their magnetic properties.

Sammanfattning

I takt med att duplexstål har blivit alltmer utbrett under de senaste åren har det blivit nödvändigt att kunna karakterisera dessa material på ett enkelt och noggrant sätt. Denna studie syftar till att förenkla och diversifiera karakteriseringsprocessen genom att fastställa samband mellan de magnetiska egenskaperna hos den ternära Fe-Cr-Ni-legeringen och dess sammansättning. För detta ändamål beräknas magnetiseringen och Curie-temperaturen för den ferritiska fasen numeriskt med hjälp av ab initio-metoder. Först beräknas grundtillståndsenergi och magnetiseringen med EMTO-metoden (exact muffin-tin orbitals). Detta ger en första uppskattning av Curie-temperaturen med hjälp av mean-field-teorin. Därefter beräknas växelverkan mellan atomerna och används för att konstruera den Heisenberg-hamiltonian som krävs för Monte Carlo-metoden. Dessa simuleringar gör det möjligt att beräkna magnetiseringen vid ändliga temperaturer och därmed bestämma Curie-temperaturen på ett andra sätt. Resultaten visar att magnetiseringen minskar när man tillsätter krom och nickel. Trenden är nästan linjär och approximeras därför väl av ett plan. De erhållna värdena stämmer överens med de som rapporterats i litteraturen. De två olika metoderna för att bestämma Curie-temperaturen ger ganska liknande resultat och visar en betydande minskning av Curie-temperaturen vid tillsats av legeringselement. Förvånansvärt nog har krom och nickel nästan identiska effekter. Resultaten ligger ganska långt från de experimentella värdena, vilket belyser begränsningarna hos Monte Carlo-simuleringar utförda i detta sammanhang. Dessa resultat visar dock den lovande potentialen hos karakterisering av duplexstål baserat på deras magnetiska egenskaper.

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Contents

1 Introduction **1**

2 Background **2**
2.1 Duplex steels 2
2.2 EMTO-CPA method 4
2.3 Thermal magnetic excitations 5
2.4 Metropolis Monte Carlo method 6

3 Method **6**
3.1 EMTO simulations of magnetization and T_C 6
3.2 Monte Carlo simulations of T_C 8

4 Results **10**
4.1 Magnetization 10
4.2 Curie Temperature 13
4.2.1 Mean-field approximation 13
4.2.2 Monte Carlo method 15

5 Discussion **19**
5.1 Practical feasibility of magnetic measurements 19
5.2 More sustainable use of steel 20

6 Conclusion **21**

7 Appendices **22**
7.1 Curie temperature chart 22

1 Introduction

Steel has shaped the recent history of humanity, its widespread use enabling the development of the modern society we know today. In 2024, approximately 1885 million tonnes of crude steel were produced worldwide [1], far exceeding the production of all other metals combined [2]. These figures can be explained by the exceptional properties of steel, which allow it to be used extensively in building and infrastructure, where 52 % of steel ends up [1]. While steel has been essential up to now, this material, and stainless steel in particular, will also play an important role in building the renewable energy system needed for the energy transition [2]. However, in its crude form, steel oxidizes and is therefore not suitable for outdoor applications, as is often required for renewable energy infrastructure. Corrosion is indeed a major problem for steel, causing a significant proportion of this material to be destroyed every year. To overcome this issue, stainless steel can be made by adding chromium. Above a weight fraction of 10.5%, this chromium will form an oxide layer that protects the iron from oxidation. These stainless steels with very attractive features can be divided into three groups, which are austenitic steels, ferritic steels and duplex steels. The last group is particularly interesting because it is composed of a mixture of austenitic and ferritic phases, which gives it exceptional mechanical properties while also providing good resistance to oxidation [3][4]. However, the desirable properties of duplex steels are closely linked to the proportion of austenite and ferrite in their composition [3]. It is generally accepted that the best properties are achieved when ferrite and austenite are present in equal proportions [5]. It is therefore important for a duplex steel manufacturer to be able to characterize its products effectively in order to guarantee the desired properties. Furthermore, in certain cases such as welding, the heating of the material leads to a redistribution of the proportions and compositions of the ferritic and austenitic phases. An effective characterization method is therefore required to ensure that the duplex steel always has a microstructure that provides the desired qualities.

There are already several suitable characterization methods available, the most robust of which is considered to be the Direct Current (DC) magnetometer [5]. This is a large machine that cannot be transported and is therefore mainly used in laboratory experiments. It is also expensive and requires a certain level of expertise to operate. A method frequently used in industry due to its ease of use and portability is the ferritoscope, which can measure the ferrite content. Although it generally produces good results, a study [5] has pointed out that cold-rolled samples were estimated to contain less ferrite than untreated samples of the same material, even though cold rolling does not alter the quantity of a phase. This method also does not allow the composition of the ferritic and austenitic phases to be determined.

In light of these challenges, this study aims to simplify the characterization process by establishing relationship between the magnetic properties of duplex steels and their composition. The focus is placed on the Fe-Cr-Ni ternary system. Using first-principles calculations, the Curie temperature (T_C) and magnetization are modelled across a range of compositions. In order to obtain a more comprehensive analysis, the Curie temperature is calculated using both the mean-field approximation and Monte Carlo simulations. This work therefore aims both to clarify the impact of the composition of duplex steels on their magnetism and to explore the possibilities offered by numerical simulations as an alternative to laboratory experiments. It also allows for a comparison

between the predictions made by mean-field theory and the widely used Monte Carlo simulations. Finally, throughout the simulation process, a number of interesting properties are calculated, providing a better understanding of the complex interactions between iron, chromium and nickel in these alloys.

2 Background

2.1 Duplex steels

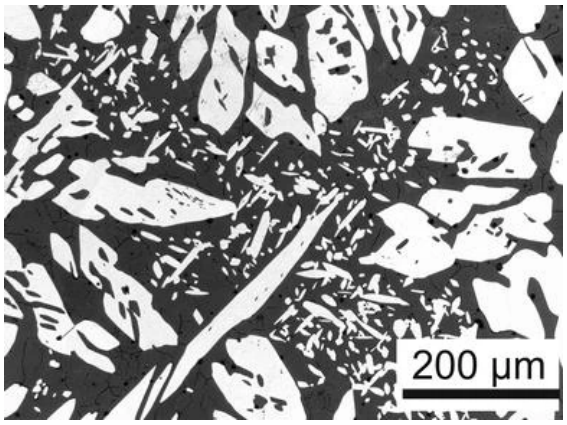
The main advantage of duplex steels lies in their biphasic composition (**figure 1a**), with each phase contributing valuables properties to the steel. Indeed, duplex steels combine high strength with high ductility and excellent corrosion resistance. They have higher strength than austenitic steels and have better impact resistance than ferritic steels. In some cases, such as with the 2205 duplex alloy, it has been reported that the material can undergo a ductile-to-brittle transition [6]. This phenomenon must be controlled so as not to compromise the steel's impact resistance. When properly manufactured, duplex steels offer excellent resistance to radiation and heat, making them suitable materials for fusion reactors [7]. During the manufacturing process, care must also be taken with regard to the carbon content of the steel. If there is too much carbon present, Cr-rich carbides may form, reducing the chromium content in both phases and making the steel more susceptible to corrosion [4][8].

Fundamentally, the coexistence of the two phases only occurs within a limited range of nickel and chromium concentrations. Two opposing effects occur in the thermodynamics of duplex steels. On the one hand, chromium, which is body-centred cubic (bcc) in crystalline form, stabilizes ferrite. On the other hand, nickel, which is face-centred cubic (fcc) in its pure state, promotes austenite. A significant addition of nickel must therefore be compensated for by adding chromium in order to maintain the presence of ferrite [3]. Conversely, steel with a high chromium content will be solely ferritic if nickel is not added. Schaeffler's diagrams for stainless steels clearly and concisely illustrate these effects (**figure 1b**). It should be noted that martensite may also be present in some cases. However, the predictions in Schaeffler's diagram should be treated with caution, as factors other than composition affect the ferrite-austenite ratio in steel. For example, the cooling rate and crystallization process affect the structure of duplex steels. It has been reported for the $Fe_{70}Cr_{20}Ni_{10}$ alloy that faster quenching produces more austenitic phase, while cold working causes the austenite to transform into the ferritic phase [9]. In addition, heat treatment such as annealing can also be applied to adjust the phase ratio [4]. A simplified description of a duplex steel can be formulated as consisting of an atomic fraction ϕ of ferrite and $1-\phi$ of austenite. In this model, the ferrite has an atomic composition of x_f chromium, y_f nickel and $1-x_f-y_f$ iron. By the same reasoning, austenite is composed of x_a atomic fraction of chromium, y_a of nickel and $1-x_a-y_a$ of iron. The system is therefore defined by five variables. The following two conservation equations can be formulated :

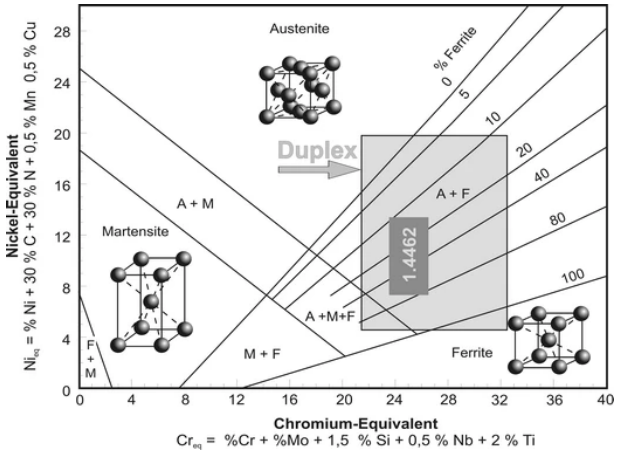
$$\phi x_f + (1-\phi)x_a = x \qquad \phi y_f + (1-\phi)y_a = y \tag{1}$$

where x is the total atomic fraction of chromium and y the total atomic fraction of nickel in the steel. Generally speaking, x and y may be known to the company that manufactured the

steel. The system therefore has three degree of freedom which, in practice, will be determined by thermodynamics and the history of the steel-making process. It would therefore be very useful to have a clear relationship between the composition of the steel and the Curie temperature, as well as another linking the composition to the magnetization, in order to provide two additional equations.



(a) Microstructure of a duplex steel showing the austenitic phase (white) and the ferritic phase (black) clearly visible [4]



(b) Schaeffler-DeLong phase diagram illustrating the crystallographic structure of the phases [4]

Some studies have already investigated the magnetic properties of certain Fe-Cr-Ni alloys, revealing extremely complex behaviour. For instance, several decades ago, some researchers [9] set out to study the magnetic phase diagram of alloys of the form $Fe_{80-x}Ni_xCr_{20}$ for x ranging from 10 to 30. They demonstrated that by increasing the amount of nickel, the system changed from a long-range antiferromagnetic system to a long-range ferromagnetic system. The transition occurs first through a spin-glass state followed by a mixture of spin-glass and ferromagnetism. Evidence of the presence of field-induced ferromagnetism has also been found. This behaviour can be explained qualitatively by the magnetic properties of the three alloy elements. Pure nickel is ferromagnetic and therefore stabilizes this phase. Chromium, on the other hand, is antiferromagnetic and promotes this behaviour. Iron also plays an important role in the magnetic properties of duplex stainless steels, as it is highly ferromagnetic in its ferritic phase. In contrast, austenite is not magnetic in pure iron, as its Curie temperature is below its onset temperature. However, it can exhibit weak magnetic properties depending on the alloying element it contains. More recently, a study [7] has numerically simulated the stability of ferromagnetic and paramagnetic phases in Fe-Cr-Ni alloys using the CPA-GPM method. They were able to obtain interesting data such as the average magnetic moment per atom. Among other things, they showed that a high concentration of chromium reduced the average magnetic moment. This can be explained by the antiferromagnetic nature of chromium, which implies a zero average magnetic moment. However, less work has been done to calculate the Curie temperature of duplex stainless steels based on their composition.

2.2 EMTO-CPA method

In this study, the EMTO-CPA method will be used for all ab initio calculations. This method is based on the reformulation of the Schrödinger equation proposed by the Density Functional Theory (DFT). This theory is very powerful because it proposed to overcome the fundamental many-electron problem, which would be numerically intractable, by considering a single electron density instead. This shifts the complexity of the previous problem into the exchange-correlation therm of the electric potential. The Hohenberg-Kohn theorem states that the energy of the system is a direct functional of the electron density :

$$E_e[n(\mathbf{r})] = T_s[n(\mathbf{r})] + E_H[n(\mathbf{r})] + E_{xc}[n(\mathbf{r})] + \int v_e(\mathbf{r})n(\mathbf{r})d\mathbf{r} \tag{2}$$

where $n(\mathbf{r})$ is the electron density as a function of the position, E_e is the total energy, T_s is the kinetic energy, E_H is the Hartree energy, E_{xc} is the exchange-correlation functional and v_e is an external potential [8]. Generally speaking, the exchange-correlation is relatively small compared to the other contributions, but it is crucial to the system's properties. Of course, $n(\mathbf{r})$ must satisfy the following normalization condition :

$$\int n(\mathbf{r})d\mathbf{r} = N \quad \text{with } N \text{ being the number of electrons} \tag{3}$$

E_{xc} is not calculated directly, as it would be too computationally intensive and must therefore be approximated using a functional. This step is important because it limits the accuracy of the DFT. In this study, the Perdew-Burke-Ernzerhof (PBE) scheme will be used, which is part of the Generalized Gradient Approximation (GGA) family. This approximation predicts results that are in closer agreement with experimental data for 3d metals (which include iron, chromium and nickel) than other methods do [8]. One way to solve these simulations would be to use a large supercell. However, this approach can quickly become challenging when modelling ternary alloys, particularly for the paramagnetic phase. The coherent potential approximation (CPA), which is directly implemented in the EMTO method [3] [8], is therefore preferred in this work.

Several key quantities are involved in ab initio methods and must therefore be defined. First, the Fermi Energy (E_F) is used in several stages of the computations. It corresponds the maximum energy associated with an occupied state. In mathematical terms, this means that if $d(E)$ is the number of states below energy level E, we have the relationship $d(E_F) = N$ [8]. Another important concept is that of the Wigner-Seitz radius (r_s), which is the basic unit of length used by the EMTO programme. First, the Wigner-Seitz cell must be constructed. It is the polyhedron formed by perpendicular bisector planes (planes perpendicular to the vector connecting two point on the lattice). For the bcc lattice, it forms a truncated octahedron as shown if the **figure 2** [10]. Next, the Wigner-Seitz radius is simply defined as the radius of a sphere whose volume is equal to the Wigner-Seitz cell :

$$\frac{4\pi(r_s)^3}{3} = \text{Volume(WS cell)} \tag{4}$$

In the case of the bcc lattice, the WS cell has a volume half that of the conventional cell. It

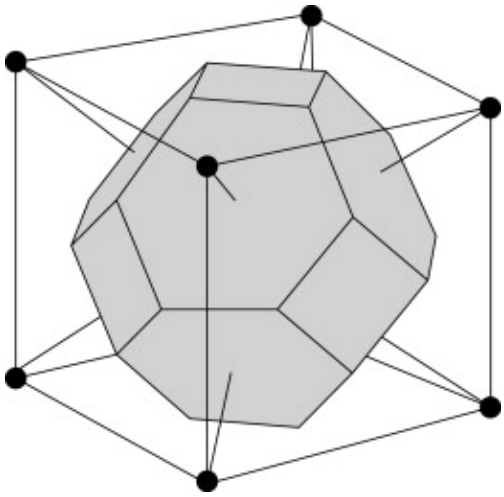


Figure 2: Wigner-Seitz cell for a body-centered cubic lattice

is therefore possible to establish a simple proportional relationship between r_s and a, which is the lattice parameter : $r_s \approx 0.492 \cdot a$.

2.3 Thermal magnetic excitations

In EMTO simulations for ferromagnetic state, all magnetic moments are assumed to point along a single common axis. As a result, each atomic site can only adopt one of two possible spin configurations: either parallel or antiparallel with respect to this axis. This collinear approximation is well-suited for describing the ground state of many magnetic systems, but it becomes increasingly inadequate as temperature rises. The disturbances that emerge in this initially ordered system are known as magnetic excitations, which can be classified into two distinct types that often occur together. The first type, referred to as longitudinal spin fluctuations (LSF), corresponds to variations in the magnitude of the local magnetic moment. The moment may therefore increase or decrease relative to its ground-state value. The second type, known as transverse spin fluctuation (TSF), involves changes in the direction of the local moment. Several studies have sought to describe these two mechanisms in detail. For example, by introducing a longitudinal spin-fluctuation Hamiltonian, a study [11] was able to model the effect of temperature on LSFs in iron and nickel. They were able to show that the behaviour of these two atoms differed significantly, partly due to the difference in their on-site energy—the energy required to vary the magnetic moment. Iron has a deep minimum in its on-site energy, which naturally gives it a large and stable local magnetic moment, even above the Curie temperature. This means that its simulated properties are in fairly good agreement with the experiment, even though the LSFs have been neglected. In contrast, nickel is much more sensitive to temperature variations, especially near T_C. The distribution of local magnetic moments is significantly altered during the transition to the paramagnetic phase. This is explained by a change in the behaviour of nickel. Below T_C, the magnetization of nickel is mainly due to the internal magnetic field. During the order-disorder transition, this internal field disappears and the behaviour of nickel is then dictated by its on-site energy.

2.4 Metropolis Monte Carlo method

The complexity of the interactions and phenomena arising from magnetic excitations demonstrates that a different method is required to calculate the magnetic moment at finite temperatures. One of the most effective techniques in this case is the Monte Carlo method and more specifically the Metropolis algorithm [12]. The first step in this algorithm is to create a supercell and assign each atom a magnetic moment with a random direction. It then calculates the energy associated with this initial configuration. Next, an atom is selected at random, and its magnetic moment is modified. The energy of this second configuration is also calculated. If the energy difference (ΔE) between the two states is negative, i.e. the system becomes more stable with this change, the second configuration is accepted. On the opposite, if $\Delta E > 0$, the change is accepted with a probability $p < 1$, where $p = exp(-\Delta E/k_B T)$ (k_B is the Boltzmann constant). This prevents the algorithm from getting stuck in a local minimum and allows it to explore configurations that become more stable after several iterations. The process is repeated many times until the algorithm reaches a termination condition.

Of course, this method also has its limitations. Firstly, it requires a way of calculating the system's energy, which is not necessarily an easy task. In the case of Monte Carlo simulations, this is generally done using the Heisenberg Hamiltonian. However, this Hamiltonian neglects longitudinal spin fluctuations, which can lead to predictions that differ significantly from experimental results. Another problem stems from the Metropolis algorithm itself. It is possible that configurations which initially increase the energy before decreasing it sharply may be overlooked. Not systematically rejecting a change that increases the energy reduces this risk but does not eliminate it, especially in the case of complex configurations.

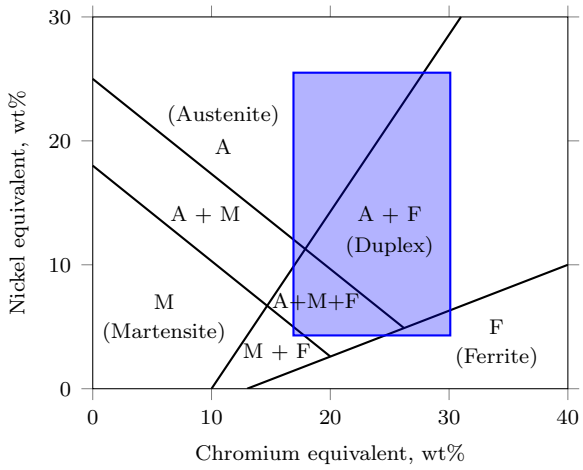
3 Method

3.1 EMTO simulations of magnetization and T_C

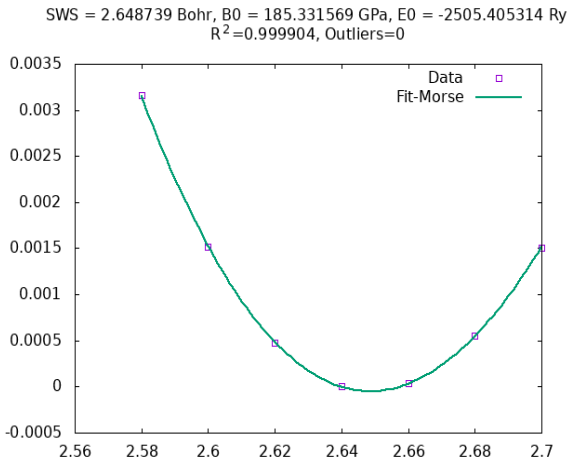
The primary focus of this study is to compute the Curie temperature and magnetization of duplex steels. To this end, simulations were performed for alloy compositions ranging from $Fe_{78}Cr_{18}Ni_4$ to $Fe_{44}Cr_{32}Ni_{24}$, with the fractions expressed as atomic percentages. This study area is shown in **figure 3a**. Although most of the alloys studied belong to the duplex family, several simulations have also been performed for austenitic and ferritic steels. This is of theoretical interest, as it allows the variation of magnetic properties to be observed across a wider range of alloy compositions. It also has a practical dimension, since depending on the manufacturing process, a mixture of both phases can be found in steels with a composition that deviates from that predicted by the Schaeffler diagram. To obtain a detailed overview of compositional effects and improve the stability of the interpolations, simulations were performed on a regular grid of compositions, with a step size of 2 at.% for each element. As a result, 88 different compositions were analysed.

In order to make the calculations feasible within a reasonable time frame, several assumptions were made regarding the structure of the duplex steel. First, a steel composed solely of ferrite and austenite was considered; that is, any presence of martensite was neglected. The ferrite phase was also simulated as being free of the intermetallic sigma (Fe-Cr). Duplex steels are considered

to be substitutional random alloys, and the energy associated with phase boundaries is not taken into account. These simplifications are based on those proposed in another study [3]. At room temperature, it is known that austenite is paramagnetic and therefore contributes little or nothing to the magnetic properties of a duplex steel. This study therefore assumes that magnetization and the Curie temperature are due solely to the properties of the bcc phase. It is therefore important to note that the composition used for the simulations is that of the ferrite (x_f, y_f) and may therefore differ from that of the alloy in general. For example, when it is stated that the simulations are performed for the $FeCr_{30}Ni_{14}$ alloy, this means that the bcc phase of the duplex steel contains 30 %at. chromium and 14 %at. nickel. The precise overall composition of the steel is, in this case, neither known nor useful for the simulations. Nor would it make sense to simulate the magnetic properties directly from the overall concentrations (x, y). Depending on the steel manufacturing process, the microstructure varies for a given concentration of chromium and nickel (x and y). This means that the distribution of alloying elements between the ferritic and austenitic phases, and therefore the magnetic properties, can differ significantly for an equivalent



(a) Schaeffler diagram [13] showing the composition area (in blue) where simulations are performed. The compositions are given here in weight percentages



(b) Energies associated with 7 different WS radii (for FeCr18Ni8). The best fit is plotted, and the square of the correlation coefficient (R^2) is shown

overall concentration. For example, using samples of 1.4462 duplex stainless steel, a study [14] applied various heat treatments to this alloy. The researchers were able to measure significant variations in the microstructure of the steel that had undergone different treatments. Steels that had been annealed at 900°C for 1 hour followed by slow cooling had a ferrite/austenite volume ratio of 0.79, whereas those that had been annealed at 1200°C for 1 hour followed by water quenching showed an increase in their ferrite/austenite ratio to 3. For these various reasons, the total content of alloying elements will not be taken into account. However, chromium is more soluble in the bcc phase than in the fcc phase. The overall composition of the alloy will therefore contain less chromium and more nickel than ferrite alone.

The simulations take place in multiple stages, with each stage providing specific information that is incorporated into the next. Initially, the bcc phase is simulated as ferromagnetic. The program simulates a crystal lattice for a specific WS radius and calculates the energy associated

with this system. To simplify the calculations, core electrons are not considered interacting directly with the electronic structure of the other atoms. However, the Schrödinger equation can no longer be used for these electrons because they have relativistic velocities, and this must be taken into account by solving the Dirac equation instead. For example, in the case of nickel, which has an electronic structure [Ar] $4s^2 3d^8$, only the electrons associated with the 3d and 4s orbitals are considered valence electrons. This process is repeated for a total of 7 different WS radii, ranging from 2.58 to 2.7 Bohr with an increment of 0.02 between each simulation. It is then possible to interpolate these 7 values to find the r_s associated with the minimum energy. To do this, the Morse equation of state is used in this work, which relates energy to the WS radius by stipulating that :

$$E(r_s) = a + be^{-\lambda r_s} + ce^{-2\lambda r_s} \tag{5}$$

where a, b, c and λ are 4 free variables which are determined so that the interpolation approximates the simulation results as closely as possible. This is illustrated in **figure 3b**, along with the stable WS radius (the one associated with minimum energy), referred to as SWS.

The same process is repeated, assuming that the bcc phase is paramagnetic. In this case, the chemical and magnetic disorder must be addressed simultaneously [3]. The program treats the alloy as consisting of six different elements, with half of the atoms having a magnetic moment oriented upward (↑) and the other half having a magnetic moment pointing downward (↓). For example, the $Fe_{70}Cr_{20}Ni_{10}$ duplex steel will be classified as an $Fe^{\uparrow}_{35}Fe^{\downarrow}_{35}Cr^{\uparrow}_{10}Cr^{\downarrow}_{10}Ni^{\uparrow}_{5}Ni^{\downarrow}_{5}$ alloy.

Once the equilibrium energies of the ferromagnetic and paramagnetic phases are known, it is possible to estimate the Curie temperature using the disordered local moment mean-field theory (DLM-MFT), which yields the following equation :

$$T_C = \frac{2}{3}\frac{E_{PM} - E_{FM}}{k_B} \quad \text{if} \quad E_{PM} - E_{FM} > 0 \tag{6}$$

where E_{PM} is the energy of the paramagnetic phase, E_{FM} the energy of the ferromagnetic phase and k_B is the Boltzmann constant [15]. If the energy of the paramagnetic phase is lower than that of the ferromagnetic phase, this means that the material will always be more stable in the paramagnetic state, and therefore $T_C = 0$. Given the equilibrium WS radius for the two simulated bcc phases, the program can calculate the magnetic moment associated with each atom in each phase. Of course, the sum of the magnetic moments must cancel out in the paramagnetic state.

3.2 Monte Carlo simulations of T_C

Now that these results have been obtained, it was possible to begin the Monte Carlo simulations. The simulations will be carried out solely for the ferromagnetic bcc phase, as this is the only phase that will have a Curie temperature. Calculating the energy associated with the paramagnetic bcc phase was useful exclusively for the mean-field approximation. The properties of this phase are no longer relevant in the subsequent calculations. A new program (still based on EMTO) calculated the exchange interaction energy between atoms. To do this, the magnetic moments associated with each atom in the ferromagnetic phase must be entered as initial parameters. Once the exchange

interaction are known, the effective Heisenberg Hamiltonian can be constructed, defined as :

$$H_{eff} = -\sum_{i \neq j} J_{ij} \mathbf{e}_i \cdot \mathbf{e}_j \tag{7}$$

where J_{ij} is the exchange interaction energy between two particular sites (i, j), and $\mathbf{e}_i$, $\mathbf{e}_j$ are unit vector pointing in the direction of local magnetic moment at sites (i, j), respectively [16]. Of course, in practice it is not possible to include interactions up to infinity. In this work, they were considered up to the 65th coordination shell. The interactions of all atoms with all others must be considered. However, the interactions are symmetric, which means that the Cr-Fe interaction must be the same as the Fe-Cr interaction and therefore does not need to be calculated. In the case of the Fe-Cr-Ni alloy, there are 6 types of interactions (Fe-Fe, Fe-Cr, Fe-Ni, Cr-Cr, Cr-Ni, Ni-Ni) to consider, resulting in a total of 390 interactions to be computed. For the remainder of the simulations, the static volume approximation was used for the exchange interactions. These are therefore calculated only once at ground state, using the equilibrium volume at zero Kelvin predicted by the EMTO method. However, as the temperature increases, the cell volume will increase, and this will have an impact on the J_{ij}. It would be more accurate to recalculate the exchange-interaction terms at each temperature and reconstruct the Hamiltonian. Unfortunately, this approach is currently difficult to implement, given that the J_{ij} are calculated by DFT, which struggles to predict properties at finite temperatures. This static volume assumption implies, amongst other things, that predictions of Curie temperatures will be more accurate when T_C is small. In some cases, it is also possible to calculate and use the J_{ij} terms associated with the paramagnetic phase. Nevertheless, in the case of the Fe-Cr-Ni ternary system, the local magnetic moments of chromium and nickel drop to zero in the paramagnetic phase. This implies that all interaction terms are zero except for the Fe-Fe one. It is therefore preferable to use the exchange interaction energies of the ferromagnetic phase. As with the energies and magnetization in the ground state, the interactions were calculated using the PBE approximation.

Once the interactions have been calculated and the Hamiltonian constructed, simulations can be carried out using the UppASD (Uppsala Atomistic Spin Dynamics) method. A simulation box containing $10 \times 10 \times 10$ primitive cell was used. 20'000 Monte Carlo steps, following the Metropolis algorithm, were carried out before the system was considered to have reached equilibrium. In this work, the magnetization was first calculated from 1 to 1'300 K, with an increment of 100 K each time. Monte Carlo simulations yield a distribution of magnetic moments per atom and the results obtained are averages of these distributions. The simulations allow us to visualize how magnetization changes with temperature. The UppASD method also calculates the magnetic susceptibility for each simulated temperature. The Curie temperature is then defined as the temperature where the change in magnetization is most abrupt (maximum of $|dM/dT|$). This coincides with the temperature where susceptibility is at its highest. An initial approximation of T_C was made, followed by further simulations every 20 K within the suspected region. The final result for the T_C calculated by the Monte Carlo method therefore has an uncertainty of $\pm$ 20 K.

4 Results

4.1 Magnetization

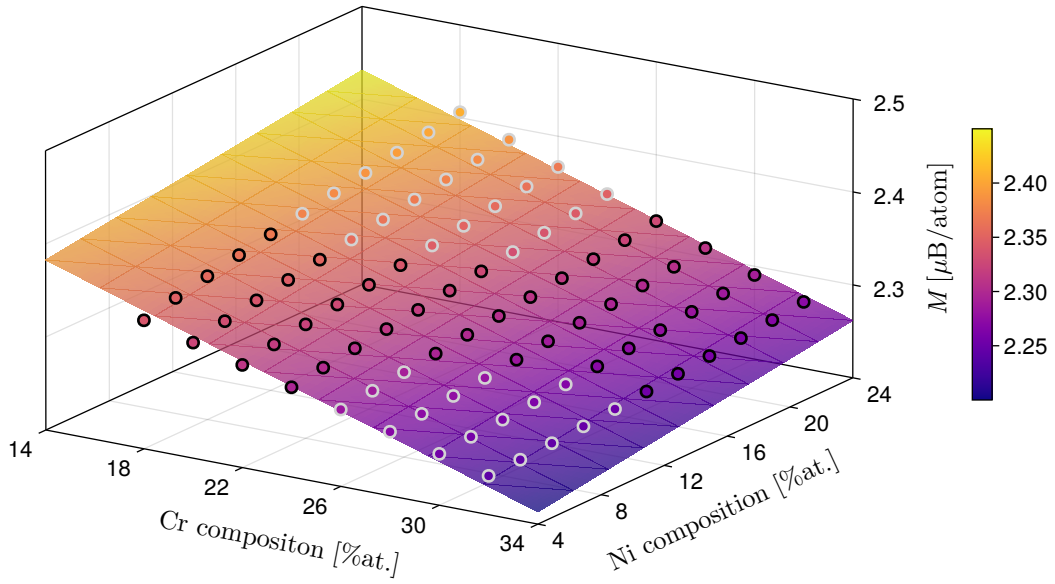


Figure 4: Magnetic moment of iron in ferromagnetic bcc Fe-Cr-Ni. Points with a light outline lie above the theoretical plane, whilst points with a dark outline lie below it.

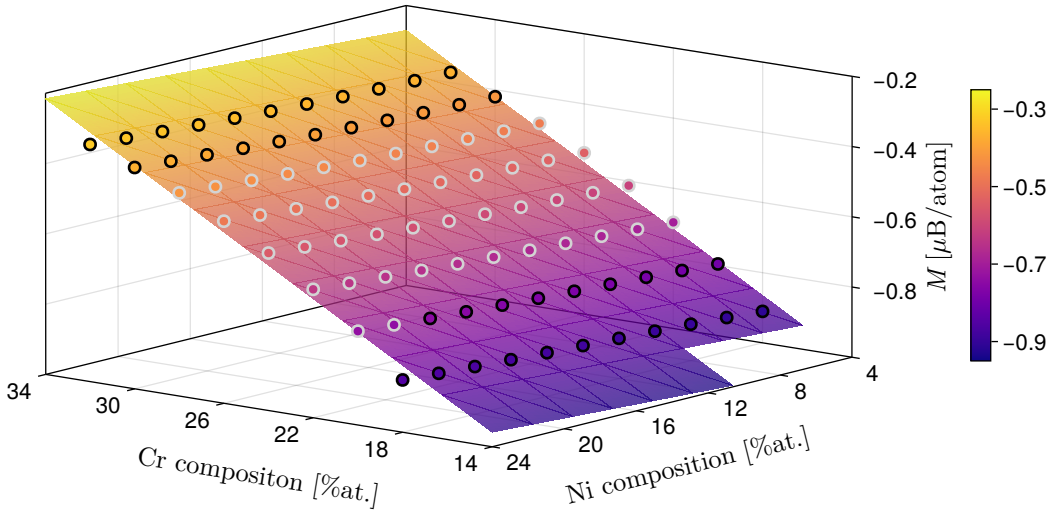


Figure 5: Magnetic moment of chromium in ferromagnetic bcc Fe-Cr-Ni. To make the data easier to read, the chromium and nickel compositions are presented differently from usual.

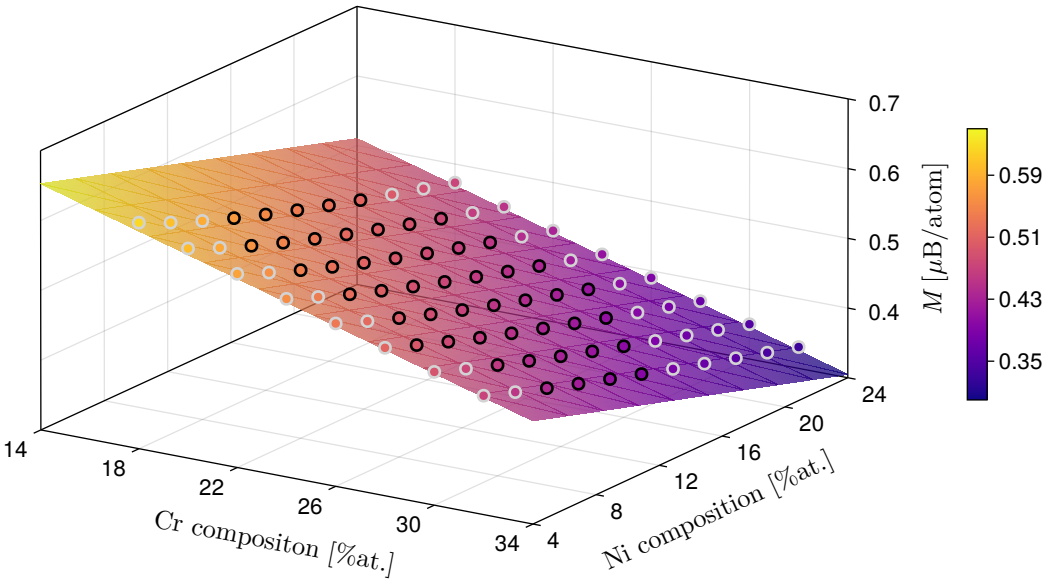


Figure 6: Magnetic moment of nickel in ferromagnetic bcc Fe-Cr-Ni

The magnetization of each atom as a function of concentration is well approximated by a plane, which can therefore provide a clearer picture of the impact of chromium and nickel on the properties. This plane can be described by the following equation :

$$M(at.\%\mathrm{Cr}, at.\%\mathrm{Ni}) = \alpha\ at.\%\mathrm{Cr} + \beta\ at.\%\mathrm{Ni} + M_0 \tag{8}$$

In the case of iron (**figure 4**), the magnetization is fairly stable and the coefficients are $\alpha = -8.5 \cdot 10^{-3}$ [μB/(atom·at.%Cr)], $\beta = 2.4 \cdot 10^{-3}$ [μB/(atom·at.%Ni)] and $M_0 = 2.4917$ [μB/atom]. This stability is to be expected for iron, given that its on-site energy has a deep minimum [11]. It is also worth noting that the magnetic moment of iron increases with nickel concentration, a finding also reported in another study [7]. The magnetic moment attributed solely to iron is reported as 2.49 μB/atom for pure iron. This is higher than the theoretical value of 2.22 μB/atom [17], which is often taken as a reference. This highlights the limitations of predicting this value using linear interpolation. Furthermore, it is not always clear which region of space, and therefore which magnetization, should be assigned to each element, particularly when dealing with atoms of different sizes. Fortunately, the magnetic moments of iron are relatively localized [18], which limits errors of this kind in this particular case.

Chromium (**figure 5**) has a negative magnetization, meaning that it points in the opposite direction to iron (**figure 4**) and nickel (**figure 6**). It therefore exhibits anti-ferromagnetic ordering relative to the other two elements. Given this information, it is easy to predict that adding chromium will reduce the average magnetization of the alloy. These interpolation coefficients are $\alpha = 37.1 \cdot 10^{-3}$ [μB/(atom·at.%Cr)], $\beta = 2.8 \cdot 10^{-3}$ [μB/(atom·at.%Ni)] and $M_0 = -1.545$ [μB/atom]. This shows that its magnetization is barely affected by the nickel content. On the other hand, chromium significantly increases its magnetization, having the highest coefficient of all the

interpolations carried out. As chromium is anti-ferromagnetic in its pure form, its magnetization is expected to be zero when the iron and nickel content is zero. It is therefore not surprising, given that M_0 is negative, that α is positive. However, the interpolation equation predicts a chromium magnetization of 2.165 μB/atom for a composition consisting solely of chromium. This highlights the limitations of the linear approximation over a wide range, particularly as it is only applied here up to 32%at. chromium. Indeed, a study [7] has shown that the magnetization of chromium in the bcc phase does not follow a linear pattern across the entire composition range.

Nickel is of interest because its magnetization decreases with increasing nickel content, which is surprising for a ferromagnetic material. This could suggest that the presence of iron constrains nickel atoms to adopt a magnetic moment larger than that observed in pure elemental nickel. However, this analysis should be taken with a degree of caution, as pure nickel crystallizes in the fcc structure whereas a bcc lattice is used in these simulations. Its magnetization is also much lower than that of iron, which is entirely to be expected. The coefficients for nickel are $\alpha = -10.3 \cdot 10^{-3}$ [μB/(atom·at.%Cr)], $\beta = -7.1 \cdot 10^{-3}$ [μB/(atom·at.%Ni)] and $M_0 = 0.825$ [μB/atom]. This trend is confirmed by another study [7], which also shows that the addition of chromium and nickel reduces the magnetic moment of nickel.

Using this information, it is possible to calculate the average magnetization by taking a weighted average of the magnetizations of each atom. This gives the equation:

$$M_{tot} = \frac{(100 - x_f - y_f)M_{Fe} + x_f M_{Cr} + y_f M_{Ni}}{100} \tag{9}$$

where x_f and y_f are given in at.% in order to be consistent with the M_{Fe}, M_{Cr} and M_{Ni}. Using the

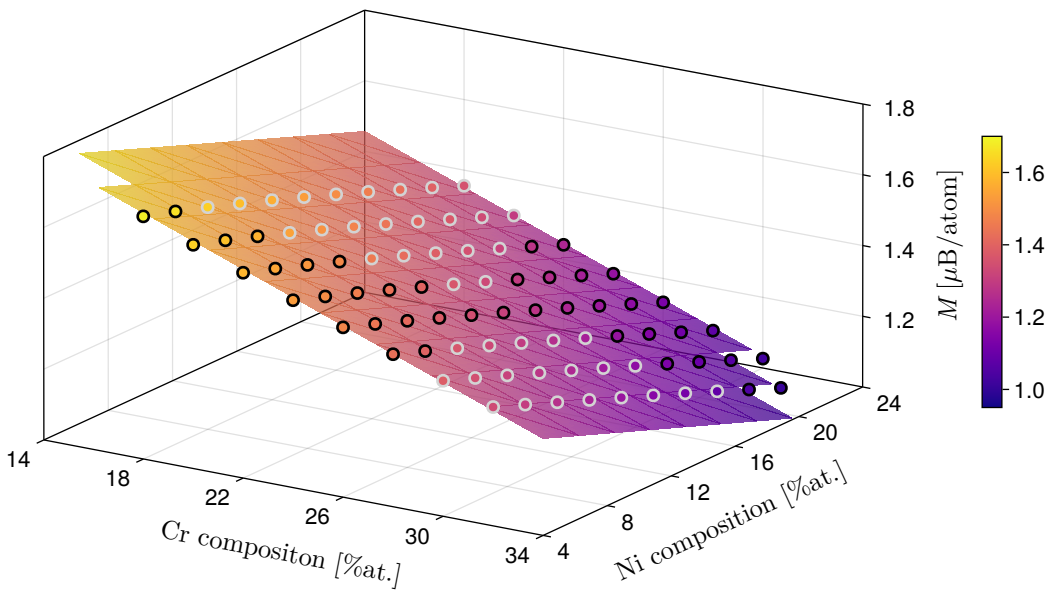


Figure 7: Total magnetization of the bcc ferromagnetic phase as a function of chromium and nickel content

equation 9 and the coefficients given for equation 8, it would therefore be possible to interpolate the total magnetization using a quadratic polynomial. However, this approach yielded a lower correlation coefficient than the linear fit did for total magnetization (**figure 7**). One reason for this is that the individual magnetization of atoms is slightly less linear than the total magnetization (the correlation coefficient is 0.9959, 0.9888 and 0.9989 for iron, chromium and nickel respectively). Thus, by choosing to interpolate using a plane, a small error is introduced three times, which accumulates. In the second case, it would appear that the tendencies to deviate from the plane cancel each other out quite effectively, making the linear approximation excellent for the total magnetization. The equation of this approximation is :

$$M(at.\%\mathrm{Cr}, at.\%\mathrm{Ni}) = -2.64 \cdot 10^{-2}\ at.\%\mathrm{Cr} - 1.72 \cdot 10^{-2}\ at.\%\mathrm{Ni} + 2.2396 \tag{10}$$

where the unit of magnetization is also given in μB/atom. The correlation coefficient is excellent in this case, standing at 0.9996. It is surprising to see such a close match between the results and the plan. However, this has been confirmed by another study [7]. This might suggest that changes in magnetization are solely due to chemical variations; in other words, that the magnetization of each atom remains constant relative to the composition, and that the total magnetization varies only because an element with a certain M is added at the expense of another. However, the magnetization of each individual atom shows that the interactions are more complex than that. Furthermore, the colour-coded representation of the points depending on whether they lie below or above the plane reveals a regular pattern in each case, suggesting that more complex functions would be better suited to describing these systems. Even using the linear approximation, the prediction for the magnetization of pure iron is excellent, yielding a theoretical value of 2.2396 μB/atom - a result very close to the reference value. If the interpolation predictions are applied to an even larger area, they indicate a magnetization of -0.4 μB/atom for pure chromium. This is much better than the previous prediction and reinforces the idea that the magnetization of the ferromagnetic bcc phase follows a trend that is close to linear across the entire range of compositions. On the whole, the results for average magnetization are consistent with the study cited in this section [7]. This is encouraging because it shows that, despite the assumptions made and outlined in the method, the results are reliable. This is an important point for the next stage, given that the ground-state magnetizations will be used as parameters for the Monte Carlo simulations.

4.2 Curie Temperature

4.2.1 Mean-field approximation

Alongside magnetization, the Curie temperature is the other magnetic property that was investigated in this work. By combining the EMTO method with mean-field theory, it is possible to obtain an approximation of T_C. The resulting function (**figure 8**) also exhibits a pronounced linear trend, making it appropriate to approximate it with a plane in order to easily visualize how the concentrations depend on the Curie temperature. In this case, the equation of the plane is the

following :

$$T_c(at.\%\mathrm{Cr}, at.\%\mathrm{Ni}) = -25.4\ at.\%\mathrm{Cr} - 23.3\ at.\%\mathrm{Ni} + 1723.7 \tag{11}$$

where temperature is expressed in Kelvin. It is interesting to note that the concentrations of nickel and chromium reduce the Curie temperature by almost the same amount. In this case too, the correlation coefficient is very high, standing at 0.9977. The pattern of the data points suggests that a parabola might provide a more accurate description of the results. However, it is not necessary to calculate its equation given the excellent agreement between T_C and a linear interpolation. In addition, the data does not span a sufficiently wide range for any curvature to be clearly distinguishable from the linear approximation, and extrapolating a second-order model beyond the data range would be meaningless, as the quadratic coefficients would not be statistically significant. It should also be noted that such a parabola would predict an even higher Curie temperature for pure iron, and thus one even further removed from the theoretical value of 1043 K. Previously, it made physical sense for magnetization to extend in the direction of chromium, as it also crystallizes in a bcc structure. However, as nickel is face-centred cubic, this means that a phase transition is to be expected between pure iron and pure nickel, implying a discontinuity in the magnetic moment. This time, the extrapolation is limited in both directions, the reason for nickel being the same as before. As for chromium, it is an antiferromagnetic element and therefore has no Curie temperature but a Néel temperature. It is therefore expected that a magnetic phase transition will occur, likely via intermediate states such as a spin glass, as has been observed in certain Fe-Cr-Ni alloys [9].

It is worth noting that mean-field theory predicts that T_C depends very strongly on the composition of the alloy. Although the difference between the predicted and experimental T_C for pure iron is very significant ($\Delta = 680$ K), it is likely that such a dependence is indeed observed. This would therefore make the Curie temperature a useful physical quantity to measure in order to characterize an alloy with good accuracy. It should be noted, however, that this measurement alone is not sufficient to predict the composition, since knowing its value merely reduces the configuration space of the bcc phase to a straight line.

Few articles discuss the Curie temperature for the bcc Fe-Cr-Ni phase in duplex steel, making direct comparisons difficult. Additionally, the Curie temperature depends solely on the composition of the ferritic phase and cannot therefore be expressed as a function of the overall alloy composition. This seems to have made regular characterization less appealing. Nevertheless, extrapolating the fitted plane allows the Curie temperature of the binary Fe-Cr system, which has been far more extensively studied, to be predicted. This makes more sense than in the case of the Fe-Ni system, as the minimum chromium concentration in the simulations is 18 %at., whereas the minimum nickel concentration is 4%at. Unfortunately, even in this case, the measurements are not necessarily taken for high concentrations. For example, one study [19] experimentally measured the Curie temperature of the Fe-Cr alloy up to 15%at. of chromium. For the maximum chromium concentration, they found a T_C of 990 K. This is also far from the mean-field prediction (using linear interpolation), which indicates a Curie temperature of 1343 K. However, it is not surprising that such a simple approximation cannot fully capture the complexity of this alloy, and it was with this in mind that the Monte Carlo simulations were carried out.

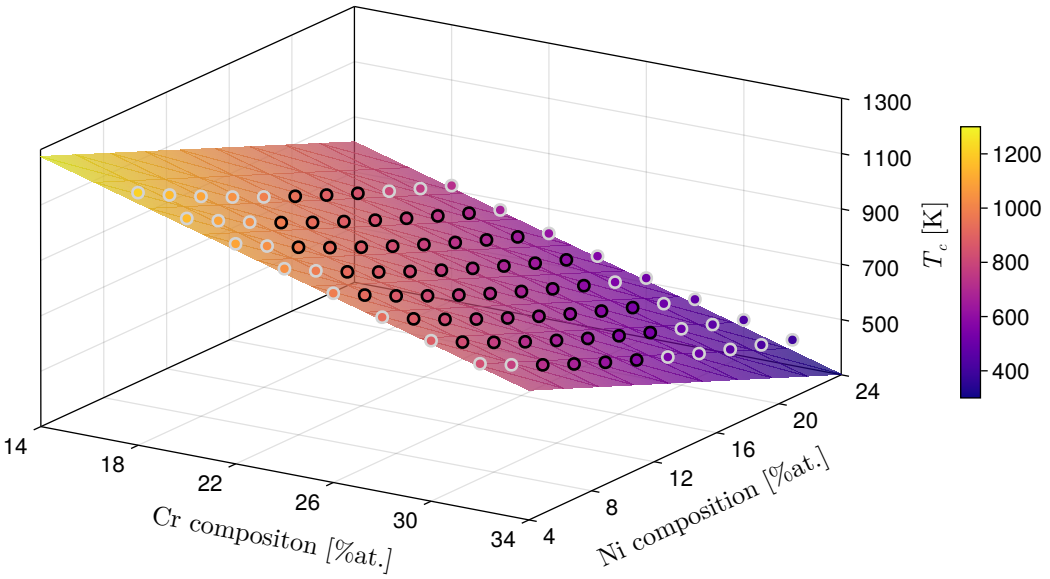
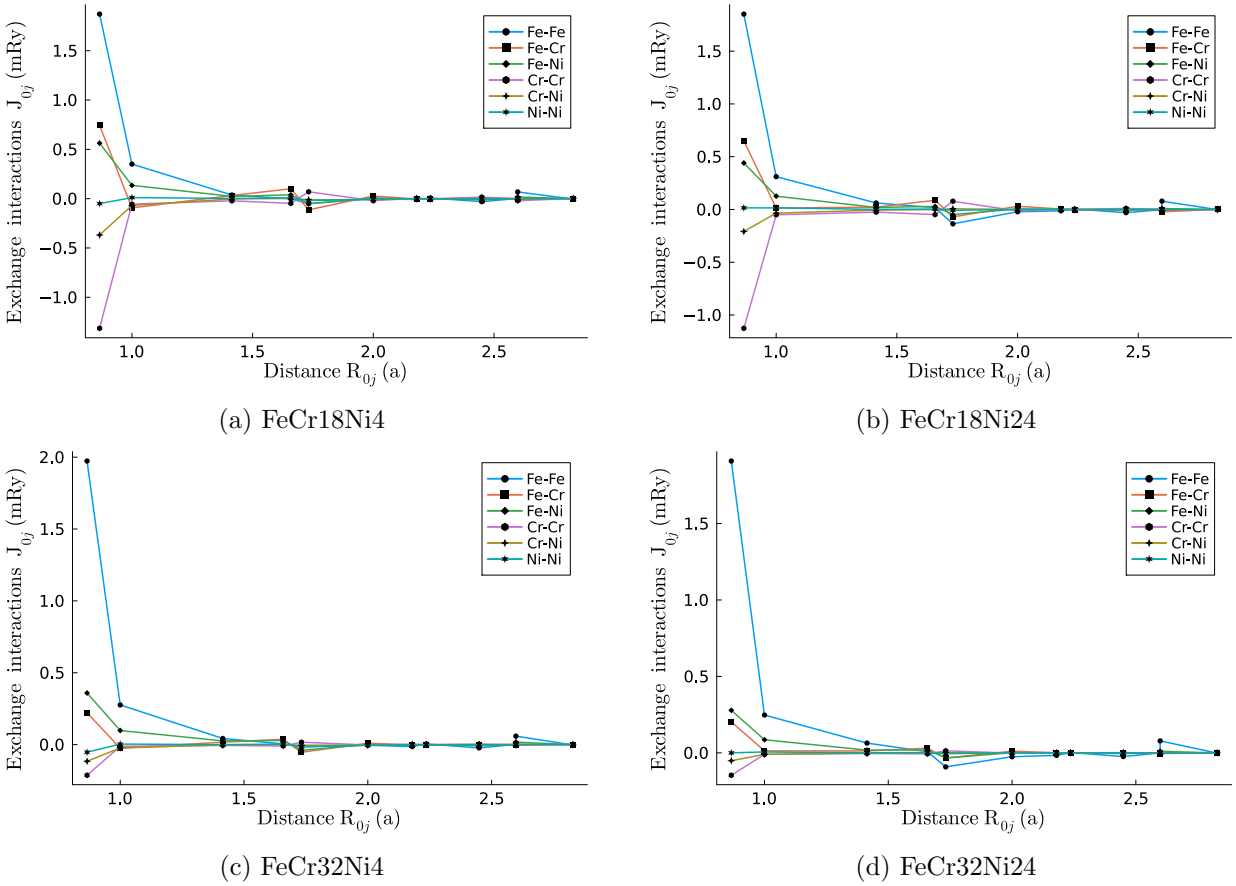


Figure 8: Curie temperature obtained from mean-field theory as a function of alloy composition

4.2.2 Monte Carlo method

Several steps are required before a prediction of the Curie temperature can be obtained using the Monte Carlo method. During this process, a number of highly interesting physical parameters are computed. Firstly, the exchange interaction energy between each atom and all the others must be calculated. These results are shown in (**figure 9**) for a selection of compositions. These compositions are each located at a corner of the region under study and therefore clearly illustrate the range of variations obtained. These results are very interesting because they reveal, more precisely than simply looking at the different magnetizations, the actual impact that each atom has on the others. As expected, iron exhibits strong ferromagnetic behaviour in all cases and a stable interaction energy regardless of the composition. The addition of chromium alone appears to slightly increase the first interaction term, as observed between the $FeCr_{18}Ni_4$ and $FeCr_{32}Ni_4$ compositions. This increase is slightly offset by the addition of nickel. Ultimately, the variation in the first Fe-Fe interaction term is virtually negligible between $FeCr_{18}Ni_4$ (1.87 mRy) and $FeCr_{32}Ni_{24}$ (1.91 mRy). Nickel, on the other hand, appears to affect the long-range order of iron, significantly reducing the exchange interaction energy of the fifth term and making it negative. It is interesting to note a peak in iron interaction at the 11th coordination shell. This is located at position a(3/2, 3/2, 3/2) in the lattice. The presence of this non-vanishing term was also noted in another study [11]. Chromium exhibits antiferromagnetic interaction with itself, which is also an expected result. However, this effect is significantly reduced by the addition of chromium, both in alloys with low and medium nickel content. This trend is particularly surprising as it suggests that, in alloys with a high chromium concentration, the Cr-Cr interaction term must be close to zero, which is more characteristic of a paramagnetic interaction than an antiferromagnetic one. Chromium



(a) FeCr18Ni4

(b) FeCr18Ni24

(c) FeCr32Ni4

(d) FeCr32Ni24

Figure 9: Interaction exchange energy for each type of pair present in duplex steel. The interactions are shown here only up to the 12th coordination shell, although they were calculated up to the 65th. The distance is given as a multiple of a, where a is the edge length of the bcc conventional cell

appears to exhibit a limited-range order, with non-negligible interactions extending up to the fifth coordination shell in the case of $FeCr_{18}Ni_4$. For the $FeCr_{32}Ni_{24}$, only the first Cr-Cr interaction term is significant, the remaining ones vanishing. The Fe-Cr interaction is another factor of significance in the bcc phase. Although the first terms are positive, they indicate antiferromagnetic interaction, as the vectors $\mathbf{e}_i$ and $\mathbf{e}_j$ were already defined as pointing in opposite directions when the Hamiltonian was constructed. In this regard, it is therefore all the more surprising to observe a negative interaction between chromium and nickel, which indicates a ferromagnetic interaction. The exchange term between iron and nickel is positive, which was also expected, as these two elements are known to align parallel to one another.

Using these exchange interaction parameters, the magnetization at a finite temperature can be calculated. The results of these simulations are shown in **figure 10** for a range of compositions. All the results show a decreasing trend, which is consistent with what is expected from Monte Carlo simulations [7]. In the experiment, however, it is expected that the magnetization will remain virtually constant at low temperatures and then decrease abruptly. This is not observed here, which indicates a certain limitation in the accuracy of Monte Carlo simulations using only

the Heisenberg Hamiltonian. It would probably be preferable to add further terms to it, taking account of longitudinal spin fluctuations, as has been done in another study [11]. An attempt was made to interpolate the magnetization as a function of temperature using the model proposed by Kuz'min [20], but this did not yield conclusive results, probably due to the slight discrepancy at low temperatures between the Monte Carlo simulations and the experimental data. It is interesting to note that the curves shown do not intersect. Ground-state magnetization therefore appears to be an important factor in determining the Curie temperature, as the lower the initial magnetization, the more rapidly it will decrease sharply, signalling the transition from the ferromagnetic to the paramagnetic phase. It is therefore unsurprising that, as the average magnetization decreases with increasing chromium and nickel content, the Curie temperature follows the same trend.

The magnetic susceptibility as a function of temperature is shown directly in **figure 11**. As these curves represent the first derivative of the magnetization with respect to temperature, they are characterized by a sharp peak at the temperature corresponding to T_C. As Monte Carlo simulations are stochastic, it has happened that simulations with the same initial parameters have produced slightly different susceptibilities, particularly around T_C where the slope is steep. Consequently, it has sometimes been difficult to determine the Curie temperature, as the peak susceptibility shifts from one temperature to the next between two simulations.

Following these various calculation steps, it is possible to plot the Curie temperature as a function of composition, as shown in **figure 12**. This time, the values are somewhat more erratic, a direct consequence of the stochastic nature of Monte Carlo simulations.

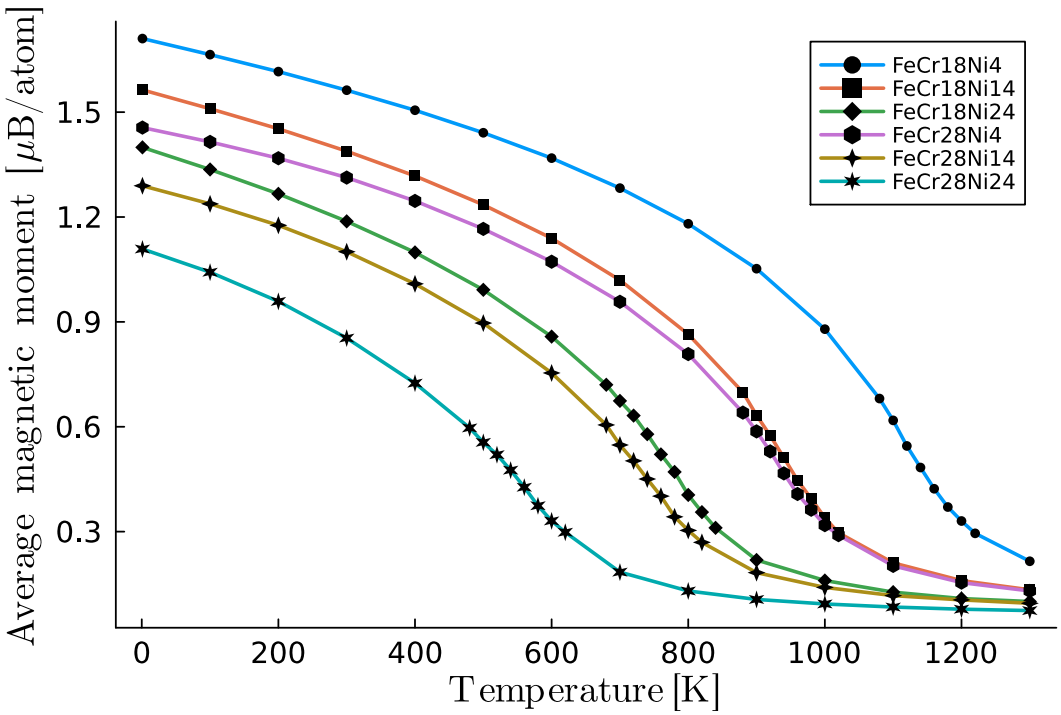


Figure 10: Magnetizations calculated using the Monte Carlo method for non-zero temperatures. The mesh is refined where M decreases rapidly

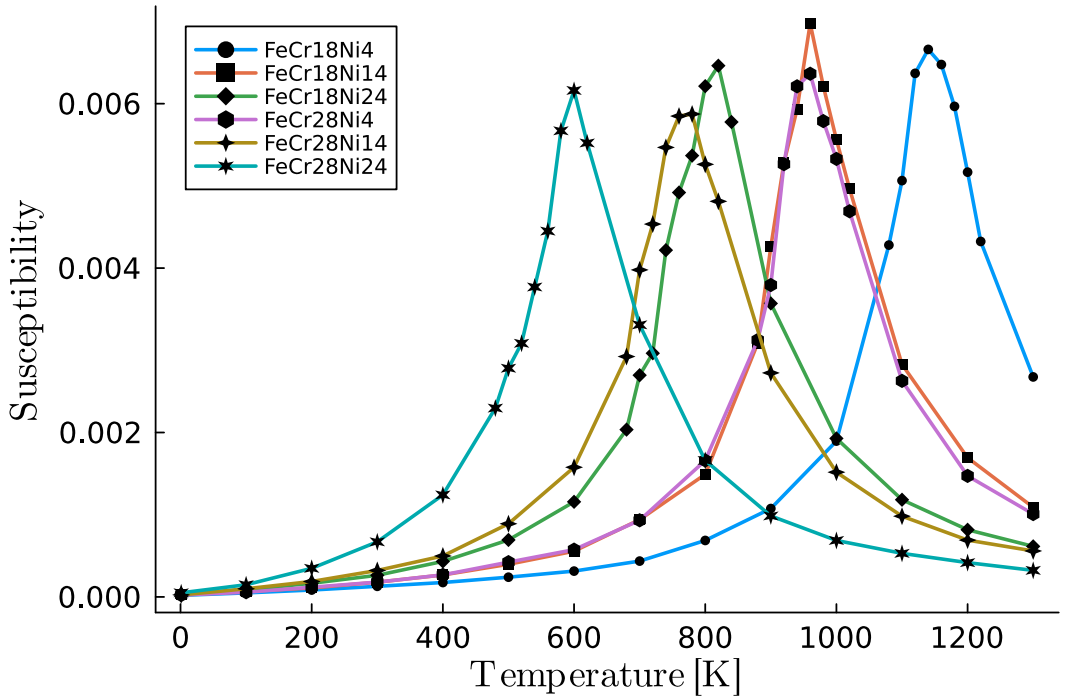


Figure 11: Magnetic susceptibility of various alloys, showing a distinct peak characteristic of the order-disorder transition. The alloy shown are the same as in the figure 10

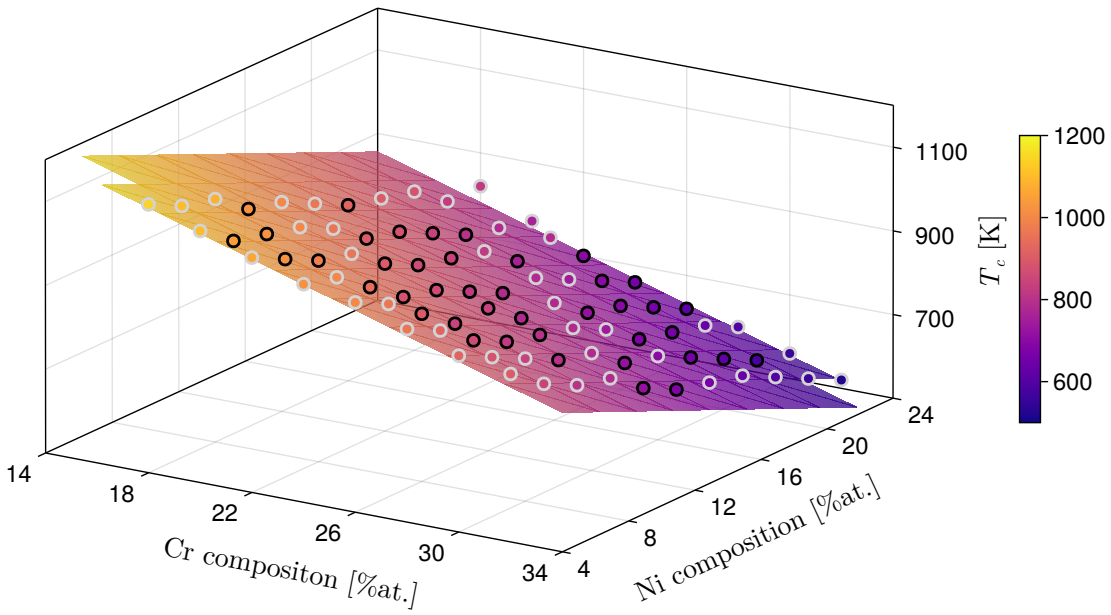


Figure 12: Curie temperature calculated using the Monte Carlo method

Moreover, the method means that susceptibility are only calculated every 20 K, which results in discrete rather than continuous variations depending on the composition. A sharp decrease in temperature as a function of the chromium and nickel content is also observed, which is consistent with the values obtained using the mean-field approximation. The data can also be accurately approximated by a plane. This plane is described by the equation :

$$T_c(at.\%\mathrm{Cr}, at.\%\mathrm{Ni}) = -18.8\ at.\%\mathrm{Cr} - 17.5\ at.\%\mathrm{Ni} + 1540.8 \tag{12}$$

As the temperatures are expressed in Kelvin, the coefficients preceding the concentrations therefore have units of $K/at.\%$. It is interesting to note that the coefficients for chromium and nickel are once again very close, indicating that the interactions within these alloys are more complex than simply ferromagnetic and antiferromagnetic couplings. The Curie temperature predicted for pure iron is again higher than the experimental value ($\Delta = 498$ K), although the difference is smaller than that obtained from the mean-field approximation. During the simulations, the magnetization at a finite temperature for pure iron was also calculated for comparison. It was then noted that the curve predicted a T_C of 820 K. This seems to indicate that the linear approximation is no longer reliable at all outside the composition range studied in this project. However, the more erratic values around the approximation plane do not allow us to predict what a better function might look like. However, the correlation coefficient remains fairly high, at 0.9948. It is therefore always worthwhile to use the linear approximation within the simulation domain. In general, mean-field theory and Monte Carlo simulations yield fairly similar results. Mean-field theory tends to amplify the extremes relative to Monte Carlo, predicting higher T_C values at the upper end and lower values at the lower end. This is well illustrated by the coefficients α and β, which are larger for mean-field theory than for Monte Carlo.

5 Discussion

5.1 Practical feasibility of magnetic measurements

In addition to seeking to understand the mechanisms governing the magnetism of duplex steels, this study also aimed to use the established relationships to characterize these alloys. As mentioned in the background section, a duplex steel, if it contains only ferrite and austenite, can be characterized by five variables with three degrees of freedom. It is these degrees of freedom that allow for such a wide range of ferrite-to-austenite ratios depending on the heat treatment applied to a given alloy. The magnetic moment is an interesting parameter because it is highly dependent on composition, which could lead to reliable predictions. Furthermore, it appears to be a parameter that can be accurately modelled using ab initio methods. However, there are also limitations to its use. First, in a practical scenario, the measured magnetization will not be the ground-state magnetization but rather the magnetization at room temperature. It has been shown that predicting magnetization at finite temperatures is a much more complicated task for numerical simulations. Moreover, it is not the magnetic moment that is directly measured in μB/atom, but rather the magnetic moment per unit volume in A/m. This means that the ferrite/austenite fraction is essential to recover the

magnetic moment per atom of the bcc phase. The magnetic moment is therefore a function of all 5 variables, and not solely of the bcc phase composition as one might have initially assumed. Intuitively, a magnetization measurement alone cannot distinguish between a large fraction of bcc phase with a small magnetic moment and a small ferrite fraction with a large magnetic moment. This clearly illustrates that all variables must be taken into account. During a magnetization measurement, a magnetic field must be applied in order to align the magnetic moments of the atoms. If this field is too large, the paramagnetic austenite could begin to contribute non-negligibly to the overall magnetism of the duplex steel. It would therefore be an error to attribute the entire response solely to the ferromagnetic phase.

The Curie temperature is the other parameter that can be measured to characterize the alloy. It is solely a function of x_f and y_f, since the austenitic phase is already paramagnetic and therefore does not undergo an order-disorder transition. In this regard, if the magnetization measurements provided a direct indication of the magnetic moment of the bcc phase, these two measurements would be sufficient to characterize the composition of the ferrite. This makes the Curie temperature an interesting measure, since prior knowledge of the chromium or nickel concentration in the ferrite would make it possible to determine the concentration of the other. There are also limitations to its application, the first being the significant discrepancy between the few experimental measurements available and the simulation results. It would be helpful, as a first step, to have a few additional reference values to determine whether the simulations differ from reality by a nearly constant factor or not. During the measurement, it is necessary to heat the alloy in order to measure its magnetism as a function of temperature. If T_C is high (around 900 K) it will be necessary to reach temperatures high enough to activate diffusion. The alloy microstructure will therefore evolve throughout the measurement, and by the end, the Curie temperature will reflect the composition of a ferritic phase that no longer corresponds to that of the original steel. This approach should therefore be preferred for alloys with relatively high chromium and nickel contents. In such cases, the wide variation in the Curie temperature with composition still makes it an interesting measurement.

Taken together, these two measurements provide two additional equations, which still leaves the system with one degree of freedom. It is therefore necessary to either make an assumption about one of the quantities, or to perform an additional measurement using one of the traditional methods. The use of a ferritoscope would be most appropriate in order to keep the measurements relatively straightforward [21]. The characterization resulting from these measurements would therefore be complete, as all variables could be determined. In addition, this would allow all measurements to be carried out without resorting to etching, which slows down the process.

5.2 More sustainable use of steel

Every year, between 25 and 33% [22] of the steel produced worldwide is destroyed by corrosion during its service life and must therefore be replaced. This naturally has a significant impact on the economy, forcing steel users – often entire countries – to pay substantial maintenance and replacement costs. Although initially less emphasized but gaining recognition in recent years, the environmental and ecological impact of corrosion now appears to be an urgent problem requiring a solution. The steel industry is already known to be polluting, having produced 10.5% of global

CO_2 emissions in 2021 [22]. Although precise data is lacking, it is estimated that 15 to 33% of the steel produced is used to replace steel destroyed by corrosion. This makes corrosion directly responsible for 1.6 to 3.4% [22] of global CO_2 emissions, possibly more than Japan (2.45% [23]). In the context of the CO_2 reductions planned under the Paris Agreement, these proportions could reach between 4.1 and 9.1% [22] of global CO_2 emissions by that time, if no action is taken in the steel sector to reduce corrosion. These figures highlight the importance of a multifaceted solution combining different approaches. It is estimated that corrosion costs could be reduced by between 14 and 33% through the adoption of best practice. This Bachelor's thesis does not, of course, provide a comprehensive solution to this problem, but by helping to simplify characterization methods, it could contribute to making the quality control of duplex stainless steels produced by industry more systematic. This would ensure that the chromium content is sufficient in both phases (austenite and ferrite) to prevent the oxidation of the steel. More generally, a more detailed understanding of the physical mechanisms involved in steel can enable the creation of steel with tailored properties. This can help reduce the amount of metal used by specifically targeting steel with the desired properties.

This project is also of interest because it offers an approach that complements traditional experimental methods by using numerical simulations to obtain data. Although they cannot replace experiments, they can help to reduce the need for laboratory work, which is generally more polluting than calculations on a computer.

6 Conclusion

This work has shed light on interesting relationships between the composition of the ferritic phase in duplex alloys and their magnetic properties, namely the magnetization and the Curie temperature. Regarding the magnetization, the results obtained using the EMTO method appear to be in good agreement with previous studies. As the Curie temperature is more challenging to compute, both the mean-field theory approach and Monte Carlo simulations were employed and compared. In both cases, the results are fairly similar, although consistently above the few experimental values found in the literature. Measuring these two magnetic properties could allow for a partial characterization of duplex steels by reducing the usual number of degrees of freedom. Although these approaches seem promising, these measurements must be conducted with caution to avoid obtaining biased results.

This broad field of research opens up several avenues for future work. As a first step, it would be worthwhile to improve the Monte Carlo simulations to obtain more accurate results. This could be achieved, for instance, by extending the Hamiltonian or by accounting for the chemical short-range order of atoms within the alloy.

7 Appendices

7.1 Curie temperature chart

Alloy	T_c MF [K]	T_c MC [K]
FeCr18Ni4	1211.3	1140.0
FeCr18Ni6	1150.9	1100.0
FeCr18Ni8	1093.9	1080.0
FeCr18Ni10	1039.6	1020.0
FeCr18Ni12	988.1	1000.0
FeCr18Ni14	938.6	960.0
FeCr18Ni16	891.0	920.0
FeCr18Ni18	845.3	900.0
FeCr18Ni20	801.2	880.0
FeCr18Ni22	758.6	820.0
FeCr18Ni24	717.5	820.0
FeCr20Ni4	1151.6	1100.0
FeCr20Ni6	1091.4	1040.0
FeCr20Ni8	1034.6	1020.0
FeCr20Ni10	980.8	1000.0
FeCr20Ni12	929.5	960.0
FeCr20Ni14	880.6	900.0
FeCr20Ni16	833.5	880.0
FeCr20Ni18	788.4	840.0
FeCr20Ni20	744.9	800.0
FeCr20Ni22	703.0	780.0
FeCr20Ni24	662.6	760.0
FeCr22Ni4	1093.5	1060.0
FeCr22Ni6	1033.5	1020.0
FeCr22Ni8	977.2	980.0
FeCr22Ni10	923.9	960.0
FeCr22Ni12	873.3	900.0
FeCr22Ni14	824.9	860.0
FeCr22Ni16	778.6	840.0
FeCr22Ni18	734.2	820.0
FeCr22Ni20	691.4	760.0
FeCr22Ni22	650.4	780.0
FeCr22Ni24	610.9	700.0
FeCr24Ni4	1037.1	1020.0
FeCr24Ni6	977.6	1000.0
FeCr24Ni8	921.9	940.0
FeCr24Ni10	869.3	880.0

Alloy	T_c MF [K]	T_c MC [K]
FeCr24Ni12	819.2	860.0
FeCr24Ni14	771.6	820.0
FeCr24Ni16	726.2	780.0
FeCr24Ni18	682.5	800.0
FeCr24Ni20	640.8	740.0
FeCr24Ni22	600.7	700.0
FeCr24Ni24	562.1	660.0
FeCr26Ni4	982.4	1000.0
FeCr26Ni6	923.6	960.0
FeCr26Ni8	868.6	900.0
FeCr26Ni10	816.7	840.0
FeCr26Ni12	767.5	840.0
FeCr26Ni14	720.8	780.0
FeCr26Ni16	676.2	780.0
FeCr26Ni18	633.4	720.0
FeCr26Ni20	592.8	700.0
FeCr26Ni22	553.6	660.0
FeCr26Ni24	516.1	620.0
FeCr28Ni4	929.5	960.0
FeCr28Ni6	871.4	920.0
FeCr28Ni8	817.3	860.0
FeCr28Ni10	766.3	820.0
FeCr28Ni12	718.1	800.0
FeCr28Ni14	672.1	780.0
FeCr28Ni16	628.4	740.0
FeCr28Ni18	586.7	680.0
FeCr28Ni20	547.0	660.0
FeCr28Ni22	509.0	640.0
FeCr28Ni24	472.6	600.0
FeCr30Ni4	878.4	920.0
FeCr30Ni6	821.2	880.0
FeCr30Ni8	768.0	840.0
FeCr30Ni10	717.9	800.0
FeCr30Ni12	670.6	780.0
FeCr30Ni14	625.6	720.0
FeCr30Ni16	582.9	700.0
FeCr30Ni18	542.3	660.0

Alloy	T_c MF [K]	T_c MC [K]
FeCr30Ni20	503.7	620.0
FeCr30Ni22	466.8	580.0
FeCr30Ni24	431.6	560.0
FeCr32Ni4	828.9	900.0
FeCr32Ni6	772.9	840.0
FeCr32Ni8	720.5	800.0
FeCr32Ni10	671.4	780.0
FeCr32Ni12	625.1	720.0
FeCr32Ni14	581.2	680.0
FeCr32Ni16	539.7	660.0
FeCr32Ni18	500.1	640.0
FeCr32Ni20	462.7	600.0
FeCr32Ni22	426.9	560.0
FeCr32Ni24	392.9	520.0

Table 1: Curie temperature for Mean-field approximation and Monte Carlo simulations

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